Artisanal Mining and Intimate Partner Violence: Evidence from Sub-Saharan Africa*

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Abstract

This paper studies the impact of artisanal mining on intimate partner violence in Sub-Saharan Africa. To identify the causal effects, we exploit spatial variation in gold suitability and industrial gold mining and exogenous temporal variation in global gold prices. Using individual-level data from 23 countries, we find that expansions in artisanal gold mining shift employment toward extraction and retail industries, with particularly strong increases in women's employment in mining, and strengthen women's intrahousehold bargaining power. These changes reduce moderately severe physical violence within the household, and the effects persist over time. The results suggest that improvements in women's economic conditions and empowerment through artisanal gold mining reduce domestic violence, with the bargaining power channel dominating potential male backlash effects.

Keywords: Gold mining, Domestic violence, Female empowerment, Intrahousehold bargaining power

JEL Codes: D12, J12, J16, O13

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1 Introduction

While industrial mining contributes substantially to Africa's exports and aggregate economic output, artisanal mining provides livelihoods for a much larger number of people. Globally, artisanal and small-scale mining directly and indirectly accounts for at least 134 million jobs (World Bank, 2020) and supports the livelihoods of 130-270 million individuals across more than 80 countries, many of which are in Africa. More than one-fourth of the global mining workforce is directly engaged in artisanal mining, around 30% of the global artisanal mining workforce consists of women (Hinton et al., 2003), compared with only 8-17% in the mining sector overall (Ellix et al., 2021). In Africa, women's participation is even higher, accounting for 40-50% of the artisanal mining workforce (ASM Inventory, 2022; Eshun, 2020). These patterns suggest that developments in this labor-intensive sector have important implications for women's economic conditions and household dynamics.

Despite this, little is known about the broader social consequences of artisanal mining within the household, particularly its effects on intimate partner violence. Intimate partner violence is one of the most serious global social concerns, with one in three women worldwide having experienced physical or sexual violence by an intimate partner during her lifetime (WHO, 2021).[1] The prevalence of domestic violence is particularly high in Sub-Saharan Africa,[2] and changes in women's economic conditions and empowerment play an important role in shaping the risk of intimate partner violence. However, the direction of the effect is theoretically ambiguous: violence may decline as women's bargaining power improves (Aizer, 2010; Anderberg et al., 2016), but it may also increase because of male backlash effects (Macmillan and Gartner, 1999; Eswaran and Malhotra, 2011). Understanding how artisanal mining affects violence within the household is therefore ultimately an empirical question, and evidence from settings with broad external validity and representative data can help inform this debate.

This paper examines the effects of artisanal and industrial gold mining on intimate partner violence in 23 countries in Sub-Saharan Africa. We show that changes in women's relative employment and income opportunities, particularly in mining and retail industries, and improvements in women's intrahousehold bargaining power operate as key mechanisms. To identify these effects, we exploit spatial variation in gold suitability and industrial gold mining, together with exogenous temporal variation in global gold prices. Conditional on industrial gold mining, higher gold prices increase the profitability of artisanal gold mining and expand local mining activity. This empirical strategy has three features that make it particularly well suited to studying the effects of artisanal

[1] Domestic violence is not only a violation of human rights but also has long-term adverse health costs, e.g., physical injury (Campbell et al., 2002), mental health disorders (Ellsberg et al., 2008), suicidal ideation (Oram et al., 2017), drug abuse (Romito et al., 2005), reproductive disorders, and sexually transmitted infections (Jewkes et al., 2010).

[2] According to the World Health Organization (WHO), the average rate of physical or sexual violence against women by husbands or partners in the past twelve months, about 20% between 2000 and 2018, was higher in Sub-Saharan Africa than in any other region.

mining in our context. First, increases in global gold prices raise the returns to extraction at both artisanal and industrial mining sites, thereby increasing the opportunity cost of non-mining activities. Higher gold prices also increase the profitability of industrial mining companies and, to some extent, the earned income of artisanal miners despite intermediaries and imperfect pass-through to local prices. Second, substantial geographic heterogeneity in gold suitability provides spatial variation in mining activity that, combined with *exogenous* temporal variation from global prices, enables us to identify causal effects. Third, combining gold suitability with the location of industrial gold mines allows us to identify the individual and interaction effects of artisanal and industrial gold mining. We first examine the effects of mining shock on different forms of domestic violence, including severe and less severe physical violence and sexual violence with different frequencies. We then evaluate several mechanisms focusing on women's intrahousehold bargaining power, gender- and industry-specific employment, household wealth, and migration.

Mining provides a particularly useful source of economic shocks to individual and household behavior in Sub-Saharan Africa, where many economies depend heavily on natural resource extraction (Thomas and Trevino, 2013; Asiamah et al., 2022). Within the mining sector, labor-intensive artisanal mining is considerably more prevalent than industrial mining in terms of both the number of workers employed and the number of sites. Among different minerals, artisanal *gold* mining (ASgM) is the dominant form of artisanal mining across Africa (Girard et al., 2025). This setting therefore provides a highly relevant context for studying how artisanal mining and the associated changes in women's economic opportunities affect violence within households.

We use nationally representative household surveys to estimate the effects of artisanal gold mining on intimate partner violence and to examine the underlying mechanisms. Specifically, we use repeated cross-sectional household- and individual-level data from the Demographic and Health Surveys (DHS), covering 23 countries in Sub-Saharan Africa. These data provide harmonized measures of intimate partner violence across countries, along with detailed individual and household characteristics. This enables us to control for determinants of domestic violence at a granular level and to examine heterogeneity across individuals, households, and countries. We combine the DHS data with highly granular grid-cell-level measures of gold suitability across Africa and with international gold prices, which vary over time and generate temporal variation in mining activity. By using nationally representative micro data and controlling for granular fixed effects, we provide population-based evidence with strong external validity while retaining rich individual- and household-level detail within and across countries.

Identifying the causal effects of artisanal gold mining on intimate partner violence is challenging because mining activity is endogenous. Reverse causality may arise if artisanal mining improves women's economic conditions and empowerment and reduces intimate partner violence, thereby making mining areas more attractive to women and further expanding mining activity.[3] In ad-

[3]Evidence from Burkina Faso shows that artisanal mining attracts young women by offering greater economic and social independence, often accompanied by changes in lifestyles and increased empowerment (Werthmann, 2009).

dition, unobserved local and demographic characteristics omitted in the regression may generate heterogeneous trends across individuals living in areas with different gold suitability, confounding estimates of mining effects. To address these concerns, we exploit spatial and temporal variation that is plausibly exogenous to household behavior and isolate the effects of industrial gold mining (ISgM) from those of geological gold suitability in order to identify the causal impact of artisanal mining (Girard et al., 2025). This strategy offers several advantages. First, using highly granular spatial information on gold suitability, we include fixed effects at the level of PRIO-GRID cells of 0.5×0.5 degrees (roughly 55×55 kilometers at the equator) as well as time-varying individual and household controls. Second, we exploit temporal variation in artisanal mining generated by changes in international gold prices, which are exogenous to local artisanal mining conditions. Potential revenues from artisanal mining depend directly on the world price of gold, yet local artisanal miners are too small to influence that price and instead take it as given in global markets (Sánchez De La Sierra, 2020). We provide indirect evidence of price pass-through from world market to local economies by estimating heterogeneous effects by internet coverage, which proxies for information flows and market connectivity. Third, identifying the effect of artisanal gold mining indirectly through gold suitability conditional on industrial gold mining enables us to capture both registered and unregistered artisanal mining activity. Finally, we introduce an additional identification strategy interacting gold suitability with industrial gold mining exposure to disentangle the effects of artisanal and industrial mining.

Our results show that moderately severe physical violence experienced sometimes within the household declines in gold-suitable areas when international gold prices rise. We document substantial labor reallocation away from agriculture and services and toward extractive activities when artisanal mining becomes more profitable. Women's employment in sales and retail activities also increases, consistent with stronger local aggregate demand and improvements in household economic conditions. These findings point to localized structural transformation following artisanal mining shocks. By contrast, conditional on artisanal gold mining activity, industrial gold mining does not significantly affect individual's employment decisions and economic activities. We also find that women's intrahousehold bargaining power increases, potentially because higher participation in mining and retail activities raises women's earned income. Indeed, employment gains in mining are stronger for women than for men. These findings suggest that, in this setting, improvements in women's bargaining power dominate potential male backlash effects. We also find evidence that separation effects may partly mediate the reduction in domestic violence following artisanal mining shocks, consistent with a theory of exposure reduction (Dugan et al., 1999). Additionally, we rule out some alternative channels related to changes in demographic composition, household wealth, or alcohol consumption. In the short run, artisanal mining has no significant effects on severe physical violence or broad measures of sexual violence. However, it increases the likelihood that women are forced into unwanted sexual acts other than sex, potentially reflecting conflict over fertility preferences. Our persistence analysis further shows that reductions in intimate partner violence, particularly moderately severe violence experienced less frequently, persist over time as women's economic participation in mining remains elevated due to sustained mining profitability. We also find that the short-run null effects on severe physical violence become negative over longer terms. In addition, initially null effects on broad measures of sexual violence become positive in the longer run. Finally, the domestic-violence-reducing effects of artisanal gold mining are concentrated in countries with unequal access to divorce, where the incidence of domestic violence is higher, as well as in middle-income countries and in West and Southern Africa, where gold suitability and industrial gold mining are more prevalent.

This study makes several contributions to the literature. First, we contribute to the literature on intimate partner violence by studying artisanal mining as a source of variation in women's economic opportunities and empowerment. Existing theoretical and empirical work suggests that improvements in women's economic conditions can have ambiguous effects on domestic violence. Better employment opportunities and higher earning potential relative to husbands strengthen women's outside options, economic agency, and intrahousehold bargaining power, which in turn reduce intimate partner violence (Farmer and Tiefenthaler, 1996; Stevenson and Wolfers, 2006; Anderberg et al., 2016). However, women's economic power may trigger male backlash effects if men seek to reassert intrahousehold authority, leading to an increase in violence within the household (Bloch and Rao, 2002; Luke and Munshi, 2011; Bobonis et al., 2013; Guarnieri and Rainer, 2021; Bhalotra et al., 2021; Field et al., 2021). Whether shocks to women's employment and income reduce or increase domestic violence is therefore ultimately an empirical question that depends on the institutional and economic context. Existing evidence focuses primarily either on unearned income shocks, such as cash transfers and dowry payments (Bloch and Rao, 2002; Angelucci, 2008; Hidrobo et al., 2016; Haushofer et al., 2019; Calvi and Keskar, 2023), or on labor demand shocks that affect earned income (Anderberg et al., 2016; Kotsadam and Villanger, 2025). Existing studies largely exploit sources of employment and income variation such as government industrial policies (Sanin, 2025), mass layoffs (Bhalotra et al., 2025), trade liberalization (Erten and Keskin, 2024), migration flows (Erten and Keskin, 2021), and randomized job offers (Kotsadam and Villanger, 2025).[4] We extend this literature by studying artisanal mining shock, a highly relevant source of variation in Sub-Saharan African economies, and by providing population-based causal evidence from a broad set of countries using nationally representative data.

Second, we add to the literature on the economic and social effects of mining. In particular, we provide the first evidence on the effects of artisanal mining on intimate partner violence. Existing research focuses primarily on large-scale, capital-intensive industrial mining at both the macro level (Sachs and Warner, 2001; Mehlum et al., 2006; Frankel, 2012) and the micro level (Berman et al., 2017; Benshaul-Tolonen, 2019), largely because of the lack of granular data on artisanal mining.[5]

[4] Employment in artisanal gold mining includes both wage work and self-employment (Goetz, 2022). Thus, changes in artisanal mining profitability may affect both labor demand and labor supply decisions.

[5] The effects of industrial gold mining on structural transformation and employment shifts across industries (Black et al., 2005; Aragón and Rud, 2013; Fafchamps et al., 2017), male and female labor supply (Santos, 2018), and intimate partner violence (Guimbeau et al., 2023) are better documented than the corresponding effects of artisanal mining. A

A small but growing literature using newly collected data on artisanal mining and mining suitability shows that artisanal mining can improve household welfare through the presence of armed actors at mines in eastern Congo (Sánchez De La Sierra, 2020), increase household consumption in Burkina Faso (Bazillier and Girard, 2020), and support local economic conditions while also generating environmental costs across Africa (Girard et al., 2025). However, little is known about its broader social consequences, particularly for women's agency, despite the substantial participation of women in artisanal mining across Africa (Eshun, 2020).

Third, we contribute to the literature on mining and violence by addressing several empirical limitations. By combining exogenous variation in global gold prices with cross-sectional variation in gold suitability and industrial gold mining exposure, our empirical strategy allows us to disentangle the effects of artisanal and industrial mining, estimate their interaction effects, and provide credible, population-based causal evidence. More broadly, a growing literature studies the relationship between extractive industries, women's welfare, and gender equality (see Baum and Benshaul-Tolonen, 2021, for a detailed review). Existing studies tend to focus either on industrial mining (e.g., Kotsadam and Tolonen, 2016; Kotsadam et al., 2017; Benshaul-Tolonen, 2024) or on artisanal mining within a specific-country context, such as the Democratic Republic of the Congo (Rustad et al., 2016), Burkina Faso (Werthmann, 2009), or India (Guimbeau et al., 2023). We further provide new evidence on the persistence of the effects of mining on individual and household behavior by examining whether current outcomes respond to international gold prices observed at different points in the past.

The rest of the paper is organized as follows. Section 2 provides background on artisanal gold mining and intimate partner violence in our setting and discusses the conceptual framework. Section 3 describes the data and summarizes descriptive statistics. Section 4 outlines the empirical strategy. Section 5 presents the main results and robustness checks, while Section 6 examines the underlying mechanisms. Finally, Section 7 concludes.

2 Background and Conceptual Framework

In this section, we first provide background on ASgM and intimate partner violence in relation to the mining activities in Sub-Saharan Africa. We then discuss theoretical frameworks to conceptually link the natural resource shock with intimate partner violence.

few recent studies examine the heterogeneous micro-level effects of industrial and artisanal mining on industry-specific employment in Sub-Saharan Africa (Girard et al., 2025), gender- and industry-specific employment in Liberia (Gräser, 2024), and violent conflict in Africa (Rigterink et al., 2025).

2.1 Artisanal Gold Mining in Africa

It is well-known that the continent of Africa is rich in mineral resources. The extraction and exploitation of mineral resources in Africa have also been increasing due to growing exploration activities in the region as there are many untapped reserves (Banerjee et al., 2015). Many African countries are highly dependent on mineral exports because of the abundance of these resources. In particular, 42 out of 54 African countries are characterized as resource-dependent, with 18 classified as dependent on non-fuel minerals. Africa produces more than 60 types of metals and mineral products, while it is assumed to have diamonds. There are several other minerals that the region hosts, including vanadium, manganese, platinum, cobalt, and gold. For example, Africa generates about 20% of the world's gold production, and Ghana and South Africa dominate the gold production within the continent (Signé and Johnson, 2021).

Although artisanal mining plays a central role in many individuals' lives and is largely prevalent in the region, the socio-economic implications of mining in Africa have been overlooked, mainly because data on mining is limited to industrial mining. Fortunately, Girard et al. (2025) propose a new measure of ASgM throughout Africa, enabling us to draw a clear and complete picture of small-scaled ASgM across the whole continent. We provide more details about the gold suitability data in the next section. As shown in Figure 1a, almost one-fifth or specifically 18% of the surface of the continent is suitable for gold and thus ASgM. In Figure 1b, we divide Africa into 10,628 grid cells of 0.5×0.5 degree or about 55×55 kilometers at the Equator (Tollefsen et al., 2012) and plot the share of each cell which overlaps with ASgM suitable bedrocks.

2.2 Intimate Partner Violence in Sub-Saharan Africa

Sub-Saharan Africa leads the world in intimate partner violence (Mossie et al., 2023). One in every five women from the region experienced physical and sexual domestic violence over the past twelve months, which is globally the highest estimate, on average, between 2000 and 2018.[6,7] Despite the high average rate, intimate partner violence incidences have declined in the region over time, primarily driven by moderately severe or less severe physical domestic violence. Severe physical and sexual intimate partner violence presented a slight downward trend from 2003 to 2019; however, these forms of violence against women in the household are generally stagnant (Figure 2a). On the other hand, the global price of gold or the gold value increased significantly after 2005, peaked in 2012, and then slightly dropped until 2019 (Figure 2b).[8] The intimate partner violence and the

[6] See the WHO's national estimates of intimate partner violence from https://vaw-data.srhr.org/data.

[7] As shown in Figure B.1, intimate partner violence incidences are concentrated in Central Africa, and incidences of mild or less severe physical violence are more prevalent than severe physical and sexual violence within the household.

[8] The surge in international gold prices between 2005 and 2012 is due to several factors, such as a greater demand from major emerging economies such as China and India (Singleton, 2014) and higher investment in commodity indices as a diversification strategy (Tang and Xiong, 2012). The commodity prices sharply increased across the board during the 2005-2008 price boom (Baffes and Haniotis, 2010). As shown in Figure B.2, the value of most minerals and crops

profitability of gold mining thus generally evolve in opposite directions: domestic violence declines while the gold price increases over time. The bivariate correlation between the incidence of any intimate partner violence in the last twelve months and the international gold price is -0.62 (SE: 0.21, p-value: 0.01) in the survey year, -0.71 (SE: 0.19, p-value: 0.00) in a year before the survey year, and -0.76 (SE: 0.17, p-value: 0.00) in two years before the survey year. Section 3 provides more details on international gold prices.

2.3 Conceptual Framework

Before discussing the theoretical concepts on the relationship between mining and intimate partner violence, it is natural to consider the theoretical links between mining and "first-stage" outcomes that are directly associated with the mining shock, such as individuals' economic activities or employment in the extractive industry, that have further implications on women's likelihood of experiencing intimate partner violence. Each woman decides the amount of her work depending on her and her husband's wage (defined as the value of their work, not necessarily the actual market wage), household income, and other factors such as time and risk preferences. As artisanal mining tends to be labor intensive and less heavy than industrial mining, local growth in mining activities spurred by the discovery of a gold mine and an increase in international gold price entails *a priori* ambiguous effects on female labor supply.[9]

First, it can trigger a substitution effect that induces more women to join the mining sector from inactivity or traditional agriculture due to the higher opportunity cost of these activities. Moreover, the local multiplier effect could reinforce female labor response through reallocation to services and trade sector (Moretti, 2010). However, such increased demand for intermediate goods and services coexisting with mining production is likely negligible for ASgM as opposed to ISgM.[10] But the indirect employment effect of ASgM could be mainly through the increased total demand. Second, increasing returns to mining employment might reduce a woman's labor supply if the income effect dominates and women's leisure is a normal good. The income effect of the gold price can also be manifested through changes in her husband's income or changes in aggregate income pooled from the household members. In addition, although women consist of 40-50% of the artisanal mining workforce (ASM Inventory, 2022), mining communities are still related to the "hyper-masculine"

sharply increased after 2005 but dropped during the 2008-2009 Great Recession. The gold price increased until 2012, as investors shifted away from speculative assets to more secure assets like gold during the uncertainty (Hergt, 2013).

[9] Our identification strategy implicitly assumes that miners are informed about the global price fluctuation to gauge the profitability from artisanal mining activities, thus miners living in regions with better information access would better benefit from the international gold price hike. We test this assumption in Section 6, where we show that the ASgM effects are more pronounced in regions with higher 3G coverage which is used as a proxy for price pass-through to the local economy.

[10] Opportunities for indirect employment in services and sales industries can increase as shown by Kotsadam and Tolonen (2016), Santos (2018) and Benshaul-Tolonen (2024) for ISgM. An empirical analysis for ASgM-induced sectoral reallocation is scant, except for Bazillier and Girard (2020), who suggested that households could take advantage of a boom in ASgM by catering to the miners' demand for goods and services.

subculture where women are substantially subject to sexual abuse by outsiders or even transactional sex (Rustad et al., 2016), thereby deterring women's labor supply through the fear channel (Siddique, 2022).[11]

Existing studies and theoretical concepts on intimate partner violence suggest that improvements in women's economic conditions and women empowerment have *a priori* ambiguous impacts on domestic violence against women. Figure 3 provides an overview of the potential mechanisms through which artisanal mining can affect intimate partner violence. On one hand, a decrease in a woman's labor supply due to the above would make her economically dependent on her husband, reducing her intrahousehold bargaining power and potentially increasing intimate partner violence. The opposite case applies if the ASgM boom increases a woman's employment opportunities and income relative to her husband, thus increasing her access to resources and expanding her outside options, which leads to a decline in domestic violence (Farmer and Tiefenthaler, 1996; Stevenson and Wolfers, 2006; Aizer, 2010; Hidrobo and Fernald, 2013; Anderberg et al., 2016). Reductions in violence may also arise mechanically through a separation effect (Dugan et al., 1999), whereby women out-migrate for work and no longer cohabit with their partners.

On the other hand, even an improvement in female job opportunities can backfire if their partners do not want them to work or it challenges the prevailing masculinity norm (Field et al., 2021; Bhalotra et al., 2021; Rustad et al., 2016). Even if husbands want their wives to work, instrumental theories of violence suggest that an increase in the probability of female employment and resulting improvement in women's bargaining power and outside options could lead to more violence from her intimate partner, who might use violence as an instrument of gaining their control over decision-making within the family (Bloch and Rao, 2002; Bobonis et al., 2013; Erten and Keskin, 2018, 2021). Domestic violence, particularly sexual violence, can also be triggered if better job opportunities for women lead them to postpone pregnancy while their husbands prefer earlier childbearing (Baizán, 2007; Bloom et al., 2009; Aaronson et al., 2021; Doepke et al., 2023).

Besides a direct change among the mining community's residence, Benshaul-Tolonen (2024) also proposes a transmission channel of a new gender perspective from female migrants, who are more open and less tolerant towards domestic violence. In contrast, an increase in husband's earnings from artisanal mining could induce higher temptation consumption, especially alcohol, which raises domestic violence (Card and Dahl, 2011; Ivandić et al., 2024; Luca et al., 2015, 2019; Basu et al., 2025). Additionally, a strong masculinity subculture of a mining community might make women more tolerant towards domestic violence and likely increase intimate partner violence. However, the association between one's attitude toward domestic violence and her experience of intimate partner violence is still unclear (Benshaul-Tolonen, 2024).

In summary, the effect of changes in employment and income opportunities due to artisanal mining on domestic violence is theoretically ambiguous and depends on which channel dominates.

[11] Sexual violence can be used strategically by the dominating armed force to gain control over valuable resources or to efficiently tax local populations on their mining production (Rustad et al., 2016; Fourati et al., 2022).

Therefore, in the empirical analysis, we examine several mechanisms that can potentially explain our findings, namely gender-specific sectoral employment, women's intrahousehold bargaining power and attitude to domestic violence, changes in household wealth, changes in the demographic composition of the mining community, alcohol consumption, and fertility decisions.

3 Data

For our empirical analysis, we combine three main data sets: Demographic and Health Surveys (DHS) for 23 countries in Sub-Saharan Africa, information on the suitability of areas for gold mining and the location of industrial gold mines covering the entire African continent, and gold prices in the global market over time. First, the DHS data provide individual-level repeated cross-section on the primary outcome variable of women's experience of different forms of intimate partner violence with different frequencies and secondary outcomes that we use to examine the potential underlying mechanisms.[12] Second, geological information on gold suitability provides spatial variation in our key explanatory variable, ASgM, when controlling for industrial gold mining. Third, time-varying information on international gold prices provides temporal variation in ASgM, enabling us to capture the evolution of the number of people involved in ASgM and the revenues they collect from such activities over time.

3.1 Intimate Partner Violence Outcomes

The main outcome we investigate in this paper is the incidence of woman's intimate partner violence from her husband or partner. The existing studies suggest that economically empowering women, e.g., via improving their employment opportunities and earning potentials, could have significant but nuanced impacts on intimate partner violence as discussed in Section 2.

The DHS data provide information on different forms of intimate partner violence experienced by the respondent or the woman, including severe physical, less severe physical, and sexual violence. We can further disaggregate these three broad forms of domestic violence. First, less severe physical violence acts include whether a woman has been (i) pushed, shaken, or had something thrown, (ii) slapped, (iii) punched with a fist or hit by something harmful, and (iv) had her arm twisted or hair pulled. Second, severe physical violence acts include whether a woman has been (i)-kicked or dragged, (ii)-strangled or burnt, and (iii)-threatened with a knife/gun or another weapon. Finally, sexual violence acts include whether a woman has been (i)-physically forced into unwanted sex, (ii)-forced into other unwanted sexual acts, and (iii)-physically forced to perform sexual acts the respondent did not want to. We also examine these detailed forms.

[12] In Appendix A.1, we provide details on our secondary outcomes, including gender- and industry-specific labor supply or economic activities, household wealth, attitude toward domestic violence, barriers to health care access, intrahousehold bargaining power, and migration.

The DHS surveys also report intimate partner violence at the extensive margin in different periods (whether a woman ever experienced domestic violence in her life, before the past twelve months, and in the last twelve months) and the intensive margin (the frequency of violence over the past twelve months: (i) sometimes and (ii) often). We focus on intimate partner violence experienced in the last twelve months at the extensive and intensive margins. The information on the frequency of domestic violence experienced by a woman in the past year allows us to investigate how ASgM shock impacts extreme and less extreme violent behaviors.

3.2 Gold Mining

We use geographic information on gold suitability throughout Africa obtained from Girard et al. (2025), who identified bedrocks suitable for gold. Like their study, we match the coordinates of the gold bedrock polygons to the PRIO-GRID 0.5×0.5 degrees cells and define dummy and continuous measures for each cell's gold suitability.[13] The dummy variable takes a value of one if a cell overlaps with a gold bedrock polygon and zero otherwise. To construct the continuous measure of gold suitability, we calculate the share of overlapping area between the cell and gold bedrock. This measure takes continuous values between zero (non-overlapping) and one (entirely overlapping), implying that a cell's gold suitability can range from totally to partially and entirely suitable.

We match these two measures of gold suitability to the DHS surveys by cells. We first determine the cell in which each DHS cluster lies and then use that information to match the gold suitability data with the DHS data at the cell level. The GPS coordinates of DHS clusters or primary sampling units (PSU) are randomly displaced by up to 2 kilometers for urban, and up to 5 kilometers for rural areas with a further displacement of 10 kilometers for 1% of the rural clusters. The displacement of DHS clusters can attenuate our estimates, but it should not bias our results as the displacement is random.

Industrial Gold Mining. Since industrial mining can also be suitable in those gold-suitable cells, it is crucial to control ISgM to isolate the impact of ASgM from that of ISgM. Hence, the effect of gold suitability will be the effect of ASgM on our outcomes after conditioning on ISgM in our regressions. For an area to be suitable for large-scale industrial mines, the concentration of gold deposits has to be high enough for the industrial mines that require a substantial investment to be profitable (Benshaul-Tolonen, 2019; Bazillier and Girard, 2020). We use data on industrial mining throughout Africa obtained from Berman et al. (2017), which comes from Raw Material Data (RMD, IntierraRMG). Active mining areas and the type of minerals are defined at the PRIO-GRID cells of 0.5×0.5 degrees between 1997 and 2010. Figure B.3 presents the distribution of industrial gold mining across PRIO-GRID cells in Africa. It shows that industrial gold mines are predominantly concentrated in the West and Southern regions of the continent.

[13] A 0.5×0.5 degrees cell is about 55×55 kilometers at the equator (Tollefsen et al., 2012).

3.3 International Gold Price

The annual international gold price over our study period comes from the World Bank Commodity Price Data (Pink Sheet). We use the gold price to construct time-varying measures of gold mining activities. In the baseline analysis, we use a one-year-lagged gold price index, which is matched to individuals based on the survey year. As robustness checks, we use the log of the lagged gold price and the current gold price as alternative measures.

3.4 Demographic and Environmental Controls

Following the relevant literature, we include several control variables in our regressions to account for other factors that affect male and female employment and intimate partner violence. Individual- and household-level demographics come from the DHS, whereas we obtain cell-level information from external sources. We discuss the cell-level factor of industrial gold mining above in this section, while Appendix A.2 provides a detailed description of the other environmental covariates we included in our regressions.

3.5 Summary Statistics

The estimation sample includes 23 countries with available DHS data on physical and sexual violence (see Table B.1 for the list of the DHS rounds and countries).[14] Table B.2 presents the descriptive statistics on intimate partner violence with different frequencies over the past twelve months. Panel A shows that about one-fourth of the women experienced domestic violence by their spouse or partner. Across various forms of domestic violence, mild or less severe physical violence is the most common, and incidences of sometimes experiencing violence are more prevalent than often experienced. In addition, Panel B further decomposes mild, severe, and sexual violence into detailed forms, and we can see that being slapped, pushed, physically forced into unwanted sex, kicked or dragged, and punched with a fist by an intimate partner are the most common. But more severe incidents, such as having their partner strangled or burnt and threatening with a knife/gun, are rare.

Table B.3 summarizes the cell-level characteristics for the 0.5×0.5 degrees PRIO-GRID cells that contain the DHS clusters. Across all DHS rounds, there are 3,948 cells. Nearly 60% of them overlap with the gold bedrock, and the mean share of overlapped area per cell is 26%. Most of them are classified as agriculturally suitable, whereas the criterion used to determine a cell's crop

[14] The sample size for three of the ten detailed forms of physical and sexual violence is slightly smaller because those questions are unavailable for up to three countries. The question on whether a woman had arm twisted or hair pulled is not available in Cote d'Ivoire. The question on whether a woman being threatened with knife/gun or other weapons is not available in Liberia. In addition, the question on whether a woman being physically forced to perform sexual acts she did not want by her husband or partner is not available in Cote d'Ivoire, Ghana, and Mozambique. Strictly restricting the sample to the 20 countries for which all questions are available yields qualitatively similar results.

suitability is derived from FAO-GAEZ.[15] Our key explanatory variable, gold suitability share $\times$ gold price, which measures a cell's time-varying potential for gold, has a mean value of 0.21 and a standard deviation of 0.30.

We present the descriptive summary for individual and household characteristics in Table B.4. The mean age for our women sample is 31.4 years. Nearly 70 percent of them live in rural areas, and more than half have a place of residence different from their place of birth. Since the sample consists of ever-married women and the DHS reports women's experience of violence from their intimate partners, the women who have experienced violence over the past twelve months could have been married or living with a partner during that period. The woman's current marital status also could differ from her marital status when she experienced domestic violence in the last twelve months. More than three-fourths of women in our sample are currently married, slightly above 10% live with a partner, and the remaining about 10% are widowed, divorced, or separated. Most women in our sample have primary education or below (78%) and are Christian or Muslim (95%). Most have also been employed in the last twelve months, particularly in agriculture, while almost all of their husbands/partners are working.

4 Identification Strategy

In this section, we describe our empirical strategy for identifying the causal impacts of artisanal gold mining on outcomes of our interest. The approach that we use is consistent with the existing studies that quantify the effect of ASgM on environmental and economic outcomes in Africa (for example, Girard et al., 2025).

4.1 Empirical Specification

To examine the causal impact of ASgM on individual-level employment decisions and intimate partner violence, we estimate the following specification:

$$\begin{aligned} Y_{ict} = \beta_0 + \beta_1 \text{Gold Suitability}_c \times \text{Gold Price}_{t-1} + \\ + \boldsymbol{X}'_{ict}\boldsymbol{\gamma} + \boldsymbol{Z}'_{hct}\boldsymbol{\theta} + \boldsymbol{M}'_{ct}\boldsymbol{\delta} + \alpha_c + \pi_{country,t} + \varepsilon_{ict}, \end{aligned} \tag{1}$$

where Y_{ict} is one of the domestic violence outcomes for woman i living in cell c at survey year t. The key explanatory variable is either a dummy or continuous variable of gold suitability for cell c interacted with the international gold price at time $t-1$, a time-varying measure of a cell's potential

[15] See Appendix A.2 for a description of the dataset used to construct the crop suitability index and how to classify a cell as agriculturally suitable.

gold value or revenue.[16] We use the lagged value of the international gold price to capture that it takes time (i) for households and individuals to adjust their activities in response to changes in the local gold price and (ii) for the world price to pass through local price. In the previous section, we discussed the individual terms of the interaction term, Gold Suitability$_c$ and Gold Price$_{t-1}$, which we do not need to include in equation (1) because we control for cell and time fixed effects as we discuss below.

The vector $\boldsymbol{X}_{ict}$ contains a set of woman i's characteristics, including age, education, marital status, and religion, a vector $\boldsymbol{Z}_{hct}$ includes a set of household characteristics including place of residence and household size and a vector $\boldsymbol{M}_{ct}$ contains a set of cell-level factors including industrial gold mining, agricultural potential, and weather conditions. Similar to the gold suitability measure, we control for an interaction between an indicator variable, set to one if the cell ever hosts an industrial gold mine during our study period, and the lagged gold price in our regressions. Controlling for ISgM measure is crucial for identifying the effect of ASgM, as the gold-suitable areas are suitable for both ASgM and ISgM. Controlling for agricultural potential, defined as the interaction between crop suitability and international crop prices, is also critical. As discussed in Section 2, commodity prices tend to move together, and thus the effect of gold price could be confounded by other price shocks. Therefore, accounting for changes in the prices of minerals and crops is important for identifying the true effect of artisanal gold mining. In Appendix A.2, we describe all cell-level variables in detail. The cell fixed effects, α_c, control for all time-invariant factors at the treatment level.

Since states or provinces and districts or communes within a country tend to be not independent of each other in Africa, i.e., each country tends to have a unified system, it is sufficient to control for country$\times$year fixed effects to capture all unobserved time-varying characteristics at the level higher than cells, such as changes in institutions, changes in labor market and domestic violence policies, the legal status of ASgM, and changes in local economic conditions. As suggested in the literature investigating the macroeconomic effects of mining, the mining boom leads to exchange rate appreciation due to increased capital inflows (Arezki et al., 2014) and an increase in the price of non-traded goods due to the income effect (Corden and Neary, 1982). These changes affect the production of different industries depending on their production input structure and goods consumption in general. Controlling for country-by-year fixed effects also captures such effects of Dutch disease, leaving the resource movement effects, i.e., employment transition across industries. Therefore, we consider that the main part of the variation left in our treatment variable is an employment shock or a shock that reallocates resources across sectors.

Given that we control for country-by-year fixed effects, we do not need to include country and time-fixed effects. The error term, ε_{ict}, captures the remaining unobserved, time-varying, and woman-specific determinants of outcomes, Y_{ict}. We cluster the standard errors at the cell level

[16] In the main results, we report the interaction between the continuous gold suitability measure and the gold price. We use a gold suitability dummy for a robustness check. The results are very robust using both measures.

to allow for heteroskedasticity and serial correlation within cells, assuming that ASgM is varied across cells.

4.2 Identification Assumptions and Interpretation

We exploit spatial variation in gold suitability and industrial gold mining, together with temporal variation in the international gold price, to identify the causal effects of artisanal and industrial gold mining. Spatial variation in gold suitability comes from differences across cells in their geological suitability for gold mining, while spatial variation in industrial gold mining comes from the location of industrial gold mines. Temporal variation comes from changes in the profitability of gold mining driven by global gold prices. A cell may be geologically suitable for gold, but gold digging or mining is more likely to occur when extraction is profitable. Interacting gold suitability and industrial mining exposure with the international gold price therefore allows us to capture changes over time in the revenue, employment opportunities, and profitability associated with artisanal and industrial gold mining.

The use of international gold prices provides plausibly exogenous time variation in local mining activity. Small-scale artisanal miners and local industrial mines are likely price takers in the global gold market, so local conditions in the cells included in our analysis are unlikely to affect the international gold price. At the same time, international gold prices are highly relevant for local mining activity because local gold prices are strongly correlated with global prices (Sánchez De La Sierra, 2020). Conditional on industrial mining exposure, higher gold prices can affect artisanal mining in gold-suitable areas through both extensive and intensive margins. On the extensive margin, an increase in the gold price makes artisanal gold mining more profitable relative to non-mining activities, encouraging entry into gold digging and exploration of new sites. On the intensive margin, higher prices induce miners to increase effort and extraction at existing artisanal mining sites.

We also account for time-varying exposure to industrial gold mining. Controlling directly for observed industrial mining activity over time may be problematic when exploiting variation in gold prices, since industrial mining may itself respond to the same price shocks and therefore constitute a "bad" control. In addition, our data on industrial gold mines are available only through 2010, while our analysis covers a longer period. We therefore construct a time-varying measure of industrial mining exposure by interacting a time-invariant indicator for whether a cell ever hosted an industrial gold mine with the international gold price. This approach captures changes in the profitability and relevance of industrial mining exposure over time without conditioning on potentially endogenous realized industrial mining activity.

Changes in global gold prices closely track changes in gold mining activity. Because we do not observe artisanal gold mining activity over time, we illustrate this relationship using the available data on industrial gold mines from 1997 to 2010. As shown in Figure 4, the number of cells hosting industrial gold mines increased steadily as the global gold price rose over time. The correlation between the global gold price and the number of cells hosting industrial gold mines is 0.96 (SE: 0.078, p-value: 0.00). In Section 6, we provide further indirect evidence that changes in world prices pass through to local prices and affect local mining activity.

The parameter of interest, β_1, in equation (1) captures the effect of artisanal gold mining rather than gold mining more broadly, because we partial out the impact of industrial gold mining in our regressions. Although this is an indirect way of identifying the effect of artisanal gold mining, its main advantage is that it captures both registered and unregistered artisanal mining. By contrast, direct measures of artisanal gold mining based only on registered artisanal mines, such as those used in Gräser (2024), may miss the predominant form of artisanal mining: unregistered mining.

Because our identification relies on cells' gold suitability and potential changes in gold-related revenue rather than actual gold production, our measure is subject to several limitations. First, the estimated impact of artisanal gold mining is likely to be attenuated by measurement error. Some cells classified as gold suitable may not contain economically viable deposits, while some cells classified as unsuitable may in fact be exposed to gold deposits because rivers can carry gold ores across space. Second, even when gold is present in a suitable cell, it may not be reachable or commercially viable. Gold deposits in some suitable cells may also be depleted over time. These sources of mismatch between measured gold suitability and actual mining activity are likely to bias our estimates toward zero. Third, additional attenuation may arise from the random displacement of DHS clusters, although such displacement should not systematically bias our estimates (Girard et al., 2025).

Another identification concern is reporting bias in survey-based measures of domestic violence, which may be correlated with changes in women's economic conditions. As women gain confidence from improved employment prospects and earnings potential, they may become more willing to report incidents of violence, potentially biasing estimated effects upward.[17] Despite this potential upward reporting bias following a positive mining shock, we find that ASgM leads to a decline in intimate partner violence, as shown in the following section. This suggests that our estimates reflect a reduction in violence even in the presence of a potential upward bias in reported violent incidents.

5 Results

In this section, we first discuss the results from estimating the effects of ASgM on intimate partner violence. We also briefly discuss the impact of ISgM. Second, we present the dynamic patterns of ASgM impacts in the medium and long term, which speaks about the persistence of the effects over time. Then, we present results from estimating heterogeneous effects. We finally examine the

[17] Relatedly, the literature examines reporting bias in survey-based measures of intimate partner violence. For example, Park et al. (2024) show that using self-interviewing methods increases reported incidents of violence by 13 percentage points in Malawi and 4 percentage points in Liberia.

robustness checks of our main results.

5.1 Average Effects

The impacts of ASgM shock on intimate partner violence experienced by women in the last twelve months are reported in Table 1. Panel A shows that the probability of women experiencing intimate partner violence, especially less severe physical violence, decreases significantly in gold-suitable zones when the gold price increases. The magnitude of the estimated coefficient suggests that a one standard deviation increase in potential profitability from ASgM activities leads to a reduction in domestic violence by 0.033. A similar calculation for the less severe physical violence shows that a one standard deviation increase in ASgM reduces the incidence of such violence by 0.028. However, the impacts of ASgM on severe physical and sexual violence are essentially zero, although the coefficient estimates are negative. Then, we split the domestic violence incidents into two different frequencies: (i) violence sometimes experienced in the last twelve months (Panel B), and (ii) violence often experienced over the past twelve months (Panel C).[18]

Most of the negative effects of ASgM on ever-experienced domestic violence over the past twelve months come from the impacts on the less frequent violence (Panel B). The results shown in Panel C suggest that any of the different forms of domestic violence often experienced in the last twelve months are not affected by the ASgM shock. This finding is intuitive because such violent behaviors remain unchanged unless the shock significantly changes men's behavior. It is also generally consistent with results from studies on alcohol controls-domestic violence relationship (for example, Basu et al., 2025) that suggest that regulating alcohol consumption is less effective in reducing the likelihood of often experiencing intimate partner violence.

Although the effects of ISgM are not our central focus, we present notable findings on ISgM effects. The impacts of ISgM are the opposite of those of ASgM and are strongly significant, except for its impact on domestic violence often experienced by women. While the effect of artisanal gold mining on the incidence of often experienced domestic violence is essentially zero, the industrial gold mining reduces incidence of such violence at the 10% significance level (Table 1). In Table B.5, we present the effects of the other two cell-level covariates on broad forms of intimate partner violence. First, crop suitability is weakly associated with an increase in domestic violence, especially sexual violence that occurred sometimes in the last twelve months. However, severe physical violence decreases in crop-suitable regions when crop prices increase. Second, temperature negatively affects intimate partner violence in general. The impacts of temperature on intimate partner violence are not statistically significant; however, it significantly reduces the incidence of severe

[18] The domestic violence-reducing impact of artisanal gold mining could be mechanical, i.e., violence reduced because employment in mining and sales requires women to spend more time outside the home away from their husbands and lower the wife's exposure to their husband. Due to the absence of data on an individual's time use, we cannot directly check this possibility. But, in the next section, we investigate other potential mechanisms, given the data availability.

physical violence (Column (3) of Panel A), particularly those that sometimes occurred in the last twelve months (Column (3) of Panel B). Like other shocks, domestic violence *often* experienced in the past twelve months does not respond to crop suitability and temperature (Panel C).

To examine the impact in more detail, we further dis-aggregate the three broad forms of domestic violence into ten forms of violence, and Table B.6 reports the estimation results. We highlight the main findings. First, we find that, among less severe physical violent activities, the likelihood of women being slapped and punched with a fist or hit by something harmful drives the violence-reducing impact of ASgM identified above (Columns (2) and (3) of Panel A). Second, similar to the results above, these effects mainly come from decreases in violence experienced sometimes (Columns (2) and (3) of Panel B), but the negative impact on the likelihood of being slapped holds for experience occurred often over the past twelve months as well (Column (2) of Panel C). Third, although the impact of ASgM shock on broadly defined sexual violence is negative but not statistically significant, the probability of women being forced into sexual acts other than sex is strongly higher in gold-suitable cells when the gold price increases (Column (9) of Panel A).

This result holds for experiences that occurred sometimes and often (Column (9) of Panels B and C). It is also generally consistent with previous findings that a rise in artisanal gold mining increases sexual violence (Fourati et al., 2022; Rustad et al., 2016; Kotsadam et al., 2017). As suggested by studies on the link between artisanal mining and sexual violence such as Rustad et al. (2016), the main reason for sexual violence against women by an intimate partner to grow in areas near artisanal mining is the subcultures of hyper-masculinity associated with mining. Additionally, we find that the probabilities of women *sometimes* being threatened with a knife/gun or other weapons (Column (7) of Panel B) and *often* being physically forced into unwanted sex (Column (8) of Panel C) decrease; however, these effects are weakly significant at the 10% level. Industrial gold mining, on the other hand, increases two types of sexual violence on which ASgM does not significantly affect. Specifically, ISgM increases the likelihood of women *sometimes* being physically forced into unwanted sex (Column (8) of Panel B) and *often* physically forced to perform sexual acts the wife did not want (Column (10) of Panel C).[19]

5.2 Persistence of the Effects

So far, in this section, we show that some forms of domestic violence in gold-suitable regions respond to a one-year lagged increase in international gold prices. But, in this part, we examine

[19]Table B.6 also shows that the positive relationship between agricultural productivity and sexual violence is driven by an increased incidence of women sometimes being physically forced to perform sexual acts the woman did not want (Column (10) of Panel B). The negative link between crop suitability and physical violence is concentrated among incidences of women having had their arms twisted or hair pulled (Column (4) of Panels A and B) and kicked or dragged (Column (5) of Panels A and B). Second, the high temperature reduces the likelihood of women being kicked or dragged (Column (5) of Panels A and B) and strangled or burnt (Column (6) of Panel B). However, it increases the probability of women being physically forced to perform sexual acts respondents did not want (Column (10) of Panels A and B).

whether those effects persist over time by estimating the domestic violence outcomes on the value of international gold price up to 10 years before the survey year, including one year before (one year lagged) like in our baseline regressions.[20]

The dynamic effects of ASgM on broad forms of domestic violence are shown in Figure B.4, and the results suggest nuanced effects that significantly evolve over time. These results include (i) the negative effect of ASgM on less severe physical violence tends to be sustained in the long run, (ii) the statistically insignificant effect of ASgM on severe physical violence turns negative and statistically significant at the 5% level in the long run, and (iii) statistically insignificant effect of ASgM on sexual violence becomes positive and statistically significant at the 10% level in the long run. The negative impact on intimate partner violence lasts only two years and becomes statistically insignificant starting from the third year after the price shock (Panel (a)). When we consider domestic violence with different frequencies, similar to our results in the previous part, the main findings for domestic violence in the last twelve months are driven by violence experienced sometimes. The only exception is that the negative impact of ASgM on intimate partner violence experienced sometimes tends to persist in the long run (Panel (b)). As presented in Panel (c), the null effects of ASgM on different forms of domestic violence experienced often remain at zero. However, the point estimates tend to turn positive despite the statistical insignificance.

For detailed forms of domestic violence, we focus on five forms of violence that the ASgM shock has a significant contemporaneous impact on, including women being (i) slapped, (ii) punched with a fist or hit by something harmful, (iii) threatened with knife/gun or other weapons, (iv) physically forced into unwanted sex, and (v) forced into other unwanted sexual acts. Figure B.5 presents the results. In terms of domestic violence shown in Panel (a), the decrease in the likelihood of women being slapped is persistent over time. However, the impacts in different periods are statistically significant at the 5% level. For the other less severe physical violence, i.e., women being punched with a fist or hit by something harmful, the weakly significant contemporaneous effect deepens in the medium term and becomes strongly significant. But it vanishes in the long run and becomes essentially zero. The significant positive impact of ASgM on women's likelihood of being forced into other unwanted sexual acts persists for about eight years, but it eventually becomes zero afterward. For intimate partner violence that happens sometimes, persistency of contemporaneous effects stays relatively longer (Panel (b)). Finally, for domestic violence that happens often, we find that those changes do not last long, except for unwanted sexual acts, which persist for about five years after the shock (Panel (c)).[21]

[20]Considering the time required for the world price to pass through to the local price and the possibility of intimate partner violence occurring at the beginning of the past year, the use of more than one year lag for the gold price, especially 2-3 years lag, also provides a robustness check of our results.

[21]As a robustness check, we also control industrial mineral mining, defined by a cell's main mineral for industrial mining activities (if any), interacting with that mineral's international price at the respective period. We also include the interaction between a cell's crop suitability and international crop price. For consistency, we use the same lag for all prices. We present the results in Figures B.6 and B.7 for broad and detailed categories of domestic violence, respectively. See Section A.2 for a description of the industrial mining dataset. The evolution of ASgM effects over

5.3 Heterogeneous Effects

We quantify heterogeneous effects within and across countries to analyze which demographic groups and country characteristics drive the average effects of ASgM on intimate partner violence. This analysis also provides initial evidence on the potential mechanisms, which are explored in greater detail in Section 6.

Within-Country Heterogeneity. First, we estimate the heterogeneous impacts in rural and urban areas. Since artisanal mining sites are more likely to be located in rural areas, we expect impacts to be concentrated among rural households. According to our data, more than 60% of rural regions are suitable for gold, while about half of the urban areas are suitable for gold. Most of the changes in the prevalence of intimate partner violence in response to ASgM shock are concentrated in rural households (Figure B.8). For example, decreases in less severe physical violence and thus domestic violence in the last twelve months in response to the mining shock that we found in our baseline estimation are concentrated among rural households (Panel (a)). The negative effects of ASgM on violence sometimes occurred in the last twelve months are also significant in the rural sample, while the effects are not statistically significant in the urban sample (Panel (b)). As shown in Panel (c), the prevalence of domestic violence often experienced by women is still not responsive for urban and rural households. The heterogeneous effects on detailed forms of violence are presented in Figure B.9. These results are intuitive and consistent with our expectations.

Second, we perform heterogeneity by women's age, which is likely a significant factor for intrahousehold bargaining power (Figure B.10). The results, shown in Panel (a), suggest that the violence-reducing effects of ASgM are relatively stronger in both magnitude and statistical significance for older women. The ASgM effects on domestic violence experienced sometimes are relatively stronger for older women (Panel (b)). The domestic violence experienced often does not change (Panel (c)), consistent with the results from heterogeneity analysis by place of residence. Figure B.11 presents the results for detailed forms of intimate partner violence.

Third, we analyze heterogeneity by household wealth since the treatment might have affected the individuals and households differently, for example, low-income individuals could be the first to choose to work on artisanal mining sites when the extractive activity becomes more profitable (Figure B.12). The negative impacts of artisanal gold mining on intimate partner violence, particularly less severe physical violence, are concentrated among households with middle wealth (Panel (a)). Panels (b) and (c) illustrate the heterogeneous impacts for intimate partner violence with different frequencies. In Section 6, we estimate the employment effects heterogeneous by household wealth to check if female employment is more pronounced than male employment among middle-wealth households.

Finally, we estimate the heterogeneous effects by the age difference between the woman and her partner, and results are shown in Figure B.13. This heterogeneity also allows us to incorporate

the 10 years is very similar to the baseline results.

men's characteristics in our analysis. We find that the violence-reducing impacts of ASgM come from women whose age differences from their husbands or partners are in the third tercile, i.e., women with age close to their partners (Panel (a)). The women in the first, second, and third terciles are approximately 18, 7, and 2 years younger than their husbands or partners, on average. The prevalence of violence from an intimate partner is higher as couples' age gets closer: the fraction of women experiencing domestic violence in any form in the last twelve months is 0.21, 0.25, and 0.27 in the first, second, and third terciles of the age difference. So, the regression results suggest that domestic violence incidents have reduced the most for women whose experience of domestic violence is more prevalent. Panels (b) and (c) show the results for domestic violence with different frequencies.

Across-Country Heterogeneity. Gender inequality in access to divorce may limit women's ability to use marital exit as an outside option, even when their economic opportunities improve (Bhalotra et al., 2021). At the same time, intimate partner violence may be more prevalent in countries where women face unequal access to divorce, because constraints on marital exit can make it more difficult for women to separate from violent partners. Changes in women's employment opportunities and earnings potential may therefore translate into intimate partner violence differently across countries depending on whether women and men have equal access to divorce. We first examine this heterogeneity in Table B.7. The results suggest that artisanal gold mining reduces the risk of intimate partner violence primarily in countries where women and men do not have equal access to divorce. These are also settings where domestic violence tends to be more prevalent: the correlation between equal access to divorce and whether a woman experienced domestic violence in the past 12 months is -0.01 (SE: 0.002, p-value: 0.01). Taken together, the results suggest that the average effect of artisanal gold mining on intimate partner violence is largely driven by countries where women face greater constraints on marital exit and where rates of domestic violence are higher. The effects of industrial gold mining display a different pattern. The positive effect of industrial gold mining on domestic violence is concentrated mainly in countries with equal access to divorce (Panels A and B). In contrast, in countries with unequal access to divorce, industrial gold mining appears to reduce women's risk of experiencing frequent domestic violence (Panel C).

Second, we estimate heterogeneous effects by country income level. Countries in our sample fall into three income groups: low-income, lower-middle-income, and upper-middle-income. We compare low-income countries with middle-income countries. Table B.8 presents the results. We find that the negative effect of artisanal mining on the likelihood of experiencing domestic violence is driven by middle-income countries, where a larger share of areas is suitable for gold mining than in low-income countries. The positive average effects of industrial mining on domestic violence identified in the baseline analysis are generally concentrated in low-income countries. In contrast, in middle-income countries, intimate partner violence declines in areas exposed to industrial gold mining when the gold price increases.

Third, we examine heterogeneity by region because, as discussed in Sections 2.1 and 3.2, gold-suitable areas and industrial gold mines are unevenly distributed across Africa. Gold-suitable areas are concentrated across much of West Africa, extend through parts of Central Africa, and are also prominent in East and Southern Africa, with dense suitability along belts running from West Africa through the Congo region and toward southeastern and southern Africa (Figure 1). Industrial gold mines are located mainly in West and Southern Africa (Figure B.3), consistent with the greater prevalence of gold-suitable areas in these regions. We therefore estimate the effects on domestic violence separately across four major African regions. As shown in Table B.9, the effects of artisanal gold mining are concentrated in West and Southern Africa.[22]

5.4 Robustness Checks

We conduct a battery of robustness checks of our main results on the concurrent effects of ASgM on intimate partner violence.

Gold Suitability Dummy. In our baseline analysis, we used the share of the cell's area suitable for gold as the spatial treatment. In the first robustness check, we replace this continuous measure of gold suitability with a dummy indicating whether a cell is generally suitable. The results, shown in Table B.12, are consistent with the baseline findings. For the detailed forms of domestic violence, statistical significance changes for a few outcomes, but the signs for those coefficients are generally similar to the main results (Table B.13).

Alternative Gold Prices. We checked the robustness of the ASgM effects in the previous part by changing the spatial portion (cell's gold suitability) of the variation in the treatment. Now, we test the robustness of our estimates by changing the time-varying part of the variation, i.e., the gold price. The baseline analysis uses the one-year lagged gold price. This section replaces the baseline price variable with either (i) the log price to account for the possibility that mining can have a concave effect on our outcomes or (ii) the current price at the survey year (Berman et al., 2017). The estimates in Table B.14 suggest that the results are remarkably robust.

The effects on the detailed forms of domestic violence also qualitatively remain the same when using these alternative gold prices, as shown in Tables B.15 and B.16.

Baseline Status of the Industrial Gold Mining. In our main analysis, we use a dummy variable indicating whether a cell ever had industrial gold mining. However, considering a potential expansion of industrial gold mines across space as the gold price increases over time, our baseline dummy could be a "bad" control. To check on this potential identification threat, we examine the robustness of the results by fixing the cell's status of industrial gold mining at the baseline or the

[22] Table B.10 presents the corresponding results for countries in East and Central Africa. In East Africa, the effects of industrial gold mining on domestic violence experienced sometimes are positive, while the effects on violence experienced often in the past 12 months are negative. The effects of industrial gold mining are not identified in Central Africa because of limited variation. The effects of artisanal gold mining on intimate partner violence are essentially zero in both East and Central Africa.

initial period when the industrial gold mining data were available for each cell. Then we interact this dummy variable with the time-varying global gold price. This interaction serves as a proxy for the intensification of industrial gold mining activities resulting from price changes in cells with industrial gold mines. Table B.17 presents the results from this robustness check on the domestic violence effects. The results are almost quantitatively the same, suggesting that the expansion of industrial gold mining in cells with no industrial gold mining activity in the initial period, due to the gold price increase, does not confound the effects of both artisanal and industrial mining on domestic violence. Only difference from the baseline regression is that the negative baseline effect of ISgM on severe physical violence often experienced by women, which was statistically significant at the 10% level, becomes statistically insignificant.[23]

Industrial Mineral Mining. The next robustness check considers replacing ISgM with industrial mining of multiple minerals (gold and other minerals, including aluminum, coal, copper, diamond, iron, lead, nickel, phosphate, platinum, silver, tantalum, tin, and zinc) since mineral prices tend to be correlated. We redefine the industrial mining variable by interacting the dummy for a cell's main mineral extracted in an industrial mine with its international price. Table B.18 reports the results. The coefficient estimates on the effect of artisanal gold mining remain qualitatively the same. The impact of ISgM on intimate partner violence that we found in our baseline analysis is also robust to an alternative measure of industrial mining, i.e., ISgM increases domestic violence towards women by their husbands and partners who are likely to gain more intrahousehold bargaining power in response to improvements in large-scale and capital-intensive industrial mining. As shown in Table B.19, the results for the detailed forms of domestic violence are also robust.

Controlling for Temperature-Induced Push Factor. While an increase in potential gold value acts as a pull factor that drives labor toward mining employment, following Girard et al. (2025), we then use changes in annual temperature as a push factor inducing the movement out of agriculture and potentially into mining due to the sector's low entry barrier (Hilson, 2016). While adverse weather conditions can have detrimental effects on agricultural production, employment, and food security, those living near the artisanal gold mines might be able to adjust their labor supply to other sectors faster, which alleviates the negative weather effect and influences the bargaining position and interaction of household members. We include an interaction between the cell's gold suitability measure and average annual temperature to capture the weather-induced push factor. Incorporating the push factor still preserves our main findings regarding the ASgM effects since the coefficient estimates on ASgM effects are remarkably similar to the baseline results. Moreover, a temperature rise in gold-suitable areas reduces the incidence of experiencing intimate partner violence, except for domestic violence often experienced by women, particularly less severe physical violence, on which the temperature in gold-suitable areas has positive impacts (Table B.20).

The domestic violence-reducing effects of temperature in gold-suitable areas are consistent with

[23] In this analysis, we treat cells with missing information on industrial gold mining in the first period as missing. The results, which are available on request, remain unchanged when we treat them as zero.

the employment shift from agriculture to mining in response to higher temperatures in gold-suitable areas. Table B.21 presents the results for detailed forms of domestic violence.

Controlling for Husband's Characteristics. In this robustness check, we include the husband's demographic characteristics, including age and education level, in our primary regression of broadly defined domestic violence outcomes. Due to missing values in the husband's characteristics, we did not control these variables in our baseline regressions but used them for robustness. Table B.22 presents the results. Although the sample size has been reduced nearly by 20%, the baseline effects of ASgM and ISgM on our primary outcomes are highly robust. The standard errors slightly increase in the regressions of domestic violence sometimes experienced in the last twelve months. However, the effects of ASgM on overall and less severe physical violence are still statistically significant, at least at the 10% level. The magnitudes of the coefficients are relatively stable. The qualitative results from regressions of domestic violence ever experienced (Panel A) and sometimes experienced remain the same.

Excluding Current Marital Status. Our baseline domestic violence regressions control for women's current marital status. However, artisanal mining and the related changes in women's employment could drive marriage market outcomes. Women might delay marriage, making the marital status or age at marriage outcomes "bad" or not valid controls. In the next section, we examine whether the artisanal mining shock affected women's current marital status as a potential mechanism. Here, however, check how much our baseline findings depend on the inclusion of current marital status as a covariate in the domestic violence regressions by dropping the variable as a control. As shown in Table B.23, the effects of artisanal mining on intimate partner violence remain unchanged.

6 Mechanisms

In this section, we examine the potential mechanisms through which ASgM could have affected intimate partner violence. We first evaluate the price pass-through or if the source of temporal variation from global gold price generates the variation in local mining activities. Second, we investigate whether ASgM improves various measures of women's intrahousehold bargaining power. Third, we estimate the effects on industry- and gender-specific employment and household wealth to evaluate the underlying reasons for changes in women's intrahousehold bargaining power. We then rule out some other potential channels, including household wealth, compositional changes, and the husband's alcohol drinking. We also provide some evidence on the separation effect or partners stopping living together as a channel for reduced physical violence and on the fertility effect as a potential channel for an increase in some forms of sexual violence.

6.1 Price Pass-Through

Our identification strategy employs international gold price as a proxy for the potential profitability of artisanal gold mining activities. It assumes that artisanal miners or intermediaries are informed about the global price movements and miners earn from international gold prices sufficient to motivate them to work on mining sites in response to price changes. Previous studies have established a strong link between international gold prices and artisanal miners' livelihood. For example, Bazillier and Girard (2020) suggest that informal gold miners in Burkina Faso earn more than 83% of the international gold price during the boom, which is much higher than the country's other popular commodities such as cotton, whose share is only 60% of the world price (van der Merwe et al., 2024). However, one might be concerned if we can empirically test to what extent the local miners benefit from an increase in international gold price and how they get information about the global market.

Despite the lack of data on local gold prices and miners' means of information access, we use the mobile network coverage as a proxy for the degree of information access to evaluate if the global price passes through to the local economy. In doing so, we estimate if the ASgM effect is stronger in cells with a 3G connection. Since individuals living in regions with 3G coverage are likely to access the internet more easily, miners and intermediaries are more likely to get informed about international price updates. We focus on 3G rather than 2G technology as artisanal miners are typically required to register for a license and sell their products to government-licensed agents or state-owned mining companies who oversee the mining sites (Hilson and Hu, 2022; Huntington and Marple-Cantrell, 2022). Those miners are paid a guaranteed share of the world market price under a price-sharing scheme (Banda and Chanda, 2021). Additionally, unlicensed and informal miners, the primary form of artisanal mining, are less likely to access the internet to check global prices. Thus, intermediaries are more likely to check international prices regularly via the internet and invest in digital technologies like 3G with better and faster internet connection. Therefore, we consider the 3G coverage is more appropriate to capture the extent of price pass-through.

We obtain data on 3G coverage for the entire African continent at 0.5×0.5 cell level from Manacorda and Tesei (2020) and define a dummy, Coverage_{ct}, which equals to one if a cell's 3G coverage is positive for a given year t. To quantify the heterogeneous effects by internet connection, we estimate the following triple interaction model:

$$\begin{aligned} Y_{ict} = \beta_0 &+ \beta_1 \text{Gold Suitability}_c \times \text{Gold Price}_{t-1} \times \text{Coverage}_{c,t-1} + \\ &+ \beta_2 \text{Gold Suitability}_c \times \text{Gold Price}_{t-1} + \beta_3 \text{Gold Suitability}_c \times \text{Coverage}_{c,t-1} + \\ &+ \beta_4 \text{Gold Price}_c \times \text{Coverage}_{c,t-1} + \beta_5 \text{Coverage}_{c,t-1} + \\ &+ \boldsymbol{X}'_{ict}\gamma + \boldsymbol{Z}'_{hct}\boldsymbol{\theta} + \boldsymbol{M}'_{ct}\boldsymbol{\delta} + \alpha_c + \pi_{country,t} + \varepsilon_{ict}, \end{aligned} \tag{2}$$

where the key explanatory variable is the triple interaction between the cell's surface suitable for gold, the lagged value of the international gold price, and the lagged dummy for a cell with 3G coverage. In this model, we use the lagged value of the internet coverage to be consistent with the lagged price, and all other terms are the same as in equation (1). The 3G coverage data from Manacorda and Tesei (2020) only span between 1998 and 2012. For 2013-2019, we fix the value of Coverage_{ct} to its 2012 value, assuming that a cell has been covered by 3G technology since then if it was covered in 2012. The parameter of interest is β_1, which captures the additional effect of ASgM on intimate partner violence in cells covered by 3G technology.

Table 2 reports the regression results. The effects of ASgM on intimate partner violence are more pronounced in cells with 3G coverage (Column (1)). In particular, the employment and income shock from artisanal mining reduces moderate or less severe physical (Column (2)) and sexual (Column (4)) violence even experienced often over the past twelve months in cells with 3G coverage. These findings support our hypothesis that the availability of 3G infrastructure plays an important role in alleviating information friction and channeling the international gold price to the local price, thereby enhancing the ASgM effects on reducing domestic violence. The economic shock from artisanal gold mining is still ineffective in reducing severely violent behavior, as we fail to find a significant impact on severe physical violence (Column (3)), similar to our baseline results.

6.2 Women's Intrahousehold Bargaining Power

Our finding on a decrease of less severe physical violence in cells more suitable for ASgM could be due to an improvement in women's intrahousehold bargaining power. To evaluate this mechanism, we estimate the effect of artisanal gold mining on various measures of intrahousehold bargaining power. The results reported in Table 3 suggest this channel is plausible. Specifically, ASgM improves women's final say in the decisions on household's large and daily purchases and family visits.[24] An improved women's bargaining power combined with a decline in intimate partner violence indicates that the bargaining power channel dominates the potential backlash effect, which is consistent with Heath and Riley (2024), who suggest no backlash effect of digital money treatment in Tanzania.

6.3 Female Employment, Structural Transformation, and Earnings Potential

We now turn to the employment effects of ASgM to better understand the mechanisms underlying the increase in women's bargaining power following artisanal mining shocks. As we discussed in Section 2, women's intrahousehold bargaining power and intimate partner violence against them depend on the wife's employment and earnings relative to the husband. Table 4 shows the estimated effects of ASgM on overall (Panel A), male (Panel B), and female (Panel C) employment in

[24] Unlike Benshaul-Tolonen (2024), we do not find any effects of ASgM on women's attitude toward domestic violence or healthcare access, as shown in Table B.24. However, we found that ISgM reduces barriers to women's medical access, although the estimated effects are statistically indistinguishable from zero.

the aggregate economy and across different industries, including mining, agriculture, services, and sales or retail. The overall employment effects generally mirror the results in Girard et al. (2025). Our innovation in employment effects of ASgM is the heterogeneous effect by gender, which is essential for our analysis. Aggregate and gender-specific labor supply decrease in response to a positive shock of ASgM; however, the negative impact is not statistically significant (Column (1)). It indicates that employment is not the main channel. But overall employment in mining or extractive activities increases due to ASgM shock, as expected, and the effect is statistically significant at the 10% level. The magnitude of the positive impact of ASgM on female labor supply is almost twice as much as that on male labor supply. The effect on female employment in mining is statistically significant at the 1% level, while the effect on male employment is statistically indifferent from zero. These indicate that women supply their labor to artisanal gold mining more than men when the industry becomes more profitable (Column (2)).

As shown in Column (3) of Table 4, in contrast, agricultural activities by men and women or male and female employment in agriculture decreases, and the effects are statistically insignificant. The magnitude of the coefficient estimates on agricultural employment effects are comparable for men and women. In Column (4) of Table 4, we show the results on employment response in services due to mining shock. The result suggests that the effects of ASgM on overall and gender-specific labor supply in the services industry are negative. The impact on male employment in services is statistically significant at the 5% level for men. It indicates a potential transition away from services, which is plausible because it could be relatively less costly to leave the services industry, particularly for men. The reason that female employment in services does not significantly decrease could be because of a positive impact of increased aggregate demand from positive mining shock. Considering that women can have a comparative advantage in services, they will be the last to leave the industry, consistent with the essentially null employment effects in services for women.

In Column (5), we find that employment in sales or retail industry increases for both men and women. Despite the statistical insignificance, the magnitude of the coefficient estimate is ten times larger for women than men. One of the reasons that women get into non-mining industries like sales/retail after the positive mining shock is that women are not allowed to go to mining sites in some countries, such as Liberia, due to traditional gender-based beliefs. This result is consistent with Gräser (2024), who show an overall pattern of women entering the sales/retail industry in response to the opening of artisanal mining. The contrasting changes in employment across industries suggest that (i) employment increases in mining and sales/retail come from agricultural and services employment and (ii) there is likely a structural transformation from non-mineral to mineral industry when mining becomes more profitable according to the Dutch disease hypothesis (van der Ploeg, 2011; Aragón and Rud, 2013). Indeed, the impact of ASgM on the probability of either a man or woman being employed is negative but not statistically significant.[25]

[25]When we estimate the gender- and industry-specific employment effects using the broader sample of 28 countries for which DHS economic activity data are available, the estimates become less noisy (Table B.25). The additional five countries are Burkina Faso, Eswatini, Guinea, Lesotho, and Madagascar. The results from the expanded sample are

Due to null overall employment impacts, the underlying mechanism for strengthening women's intrahousehold bargaining power is less likely to be higher overall employment, at least on the extensive margin. However, our results suggest that changes in sector-specific employment or structural transformation play a role in explaining the response of intrahousehold dynamics to artisanal mining in our context. So, we further explore other potential mechanisms. Several mechanisms could have increased a woman's bargaining power within the household, such as an increase in her earnings relative to her husband or partner. As theoretically modeled in Basu (2006), a woman's bargaining power or "balance of power" can be influenced by the household's decision, for example, through what she earns rather than the female wage rate or employment status. First, female employment in mining and sales industries increased, which could have led to a rise in women's earnings.[26] Second, although we cannot empirically check due to the data limitation, women's hours and days spent on mining activities and retail work could have intensively increased, leading to a rise in their total earnings. Third, male employment does not increase significantly in any industry and declines in services, whereas female employment rises significantly in mining and does not decline in any sector. Thus, earning potential might have improved more for women than men, strengthening women's intrahousehold bargaining power relative to men.

The lack of data on individual-level income restricts this paper from directly checking if the wife's earnings increased more than her husband's. However, we estimate the effect of ASgM on household wealth, combined with the employment effects above, to indirectly examine a channel through which a woman's income increases relative to her husband or partner. The effect of ASgM on household wealth in Sub-Saharan Africa has been estimated in Girard et al. (2025); however, we re-estimate the wealth effects under five different scenarios to offer some validity and robustness. We document a significantly positive impact of ASgM on household wealth (Table 5). We cannot disentangle the contribution of the wife and husband to the household wealth growth. However, we consider that the wife's income increases more because (i) household wealth increases and (ii) the sector-specific positive employment effects are more significant in magnitude and statistical significance. Thus, the household wealth-increasing impact of ASgM and our results on employment responses indicate that the increase in payrolls or income from extractive activities was substantial, especially for women. An increase in overall and female employment in the sales or retail industry, shown in Column (5) of Table 4, could also indicate an improvement in aggregate demand through higher income.

As discussed in the previous section, we estimate the employment effects heterogeneous by household wealth. Figure B.14 shows the results. An increase in mining employment is higher for low-wealth households, and the positive impact on mining employment by women is strongly significant for low- and middle-wealth households. Female labor supply strongly increased even among middle-wealth households (Panel (c)), while men's mining employment weakly increases

driven in part by Guinea, for which four DHS rounds are available and therefore provide greater temporal variation.

[26]The DHS does not report an individual's wage and earnings, so we cannot examine how ASgM affects women's earnings relative to men. But, we have shown that household wealth increased due to the ASgM.

only in low-wealth households (Panel (b)). This result supports our findings above on the primary mechanism, indirectly suggesting that higher female earnings from more female employment in the mining industry relative to men leads to a decline in intimate partner violence.

6.4 Separating the Effects of Artisanal and Industrial Gold Mining

We first check the robustness of the household wealth effect of artisanal gold mining using the cell's *baseline* status of industrial gold mines instead of the cell's status of *ever* hosting ISgM (Columns (1) and (2) of Table B.26). The effect of ASgM on household wealth is almost quantitatively the same, while the significant effect of ISgM disappears, which is consistent with Girard et al. (2025). This intuitively indicates that artisanal mining is more inclusive than industrial mining. When we fix the ISgM status at the baseline period, the effect of ASgM could be biased, as the gold suitability might capture the effect of the expansion of ISgM in cells that did not have an industrial gold mine at the baseline period due to an increase in gold prices. To investigate this, we drop the ISgM measure altogether and find that the coefficient estimate on the gold suitability in the household wealth regression is remarkably unchanged (Column (3) of Table B.26). This questions our identification strategy and suggests that we might not be identifying the effect of artisanal gold mining by partialling out industrial gold mining from gold suitability.

Thus, to check if the gold suitability captures the effect of artisanal gold mining, we estimate the following model to estimate the heterogeneous effect of industrial gold mining and artisanal gold mining in gold-suitable areas in response to an increase in gold price:

$$\begin{aligned} Y_{ict} = \beta_0 &+ \beta_1 \text{Gold Suitability}_c \times \text{ISgM}_c \times \text{Gold Price}_{t-1} + \\ &+ \beta_2 \text{Gold Suitability}_c \times \text{Gold Price}_{t-1} + \text{ISgM}_c \times \text{Gold Price}_{t-1} + \\ &+ \boldsymbol{X}'_{ict}\gamma + \boldsymbol{Z}'_{hct}\boldsymbol{\theta} + \boldsymbol{M}'_{ct}\boldsymbol{\delta} + \alpha_c + \pi_{country,t} + \varepsilon_{ict}, \end{aligned} \tag{3}$$

where ISgM_c is our baseline measure of industrial gold mining, an indicator variable that takes a value of one if the cell ever hosts an industrial gold mine during our study period, and other terms are similar to those in equation (1).

Table B.27 presents the effects on the household wealth using alternative measures of ISgM_c dummy. The results suggest that both artisanal and industrial gold mining in gold-suitable areas increase household wealth when the gold price rises. In terms of both magnitude and statistical significance, industrial gold mining in areas suitable for gold increases household wealth more than artisanal gold mining. Industrial mining in non-suitable areas does not benefit households in terms of wealth or income. Then, in Table B.28, we show the effects of artisanal and industrial mining on broad forms of domestic violence. The results suggest that artisanal gold mining in gold-suitable areas reduces the women's risk of experiencing violence within the household, while industrial gold mining in gold-suitable areas has no effect on domestic violence despite its strong household wealth effect. The changes in domestic violence are driven by those experienced *sometimes* in the last twelve months. These confirm that our baseline effect of artisanal gold mining on domestic violence captured by the effect of gold suitability is indeed the effect of artisanal gold mining.

6.5 Household Wealth

We further check whether a reduction in intimate partner violence is mediated through a change in household wealth because Haushofer et al. (2019) show that changes in household income affect the intimate partner violence in the context of a cash transfer program in Kenya. Thus, we run the baseline regression on domestic violence outcomes with household wealth index as an additional covariate, and we report the results in Table B.29. Higher household wealth is associated with a lower incidence of domestic violence. However, since the magnitude of ASgM coefficients hardly changes, except for a slight decrease, household wealth is not the primary channel through which ASgM affects intimate partner violence. The findings are the same for detailed forms of domestic violence (Table B.30).

6.6 Composition Changes, Separation, and Migration

We check if the baseline results reflect the changes in the composition of the mining communities over time in response to the gold price shock. For example, a boom in ASgM could trigger more inflows of migrants, particularly young women, to the communities, and those women carry different gender attitudes that help to transmit to the host communities (Werthmann, 2009; Benshaul-Tolonen, 2024; Kotsadam and Tolonen, 2016). They are also less likely to engage in traditional agriculture and more likely in mining or services/trade activities. We, therefore, regress women's characteristics on the explanatory variable controlling for the same cell-level covariates and fixed effects, and Table B.31 reports the results. We found no significant impact of ASgM on all characteristics except that, after the gold price increase, women living in gold-suitable areas are more likely to be separated from a consensual union (Column (5)). It could happen in two cases. First, the number of couples who are separated or not living together could have increased if more women in such situations migrated into the area. Second, couples who used to live together start to live separately when the economic and employment opportunities rise due to a gold price hike.

To evaluate the first possibility, we examine the women's migration pattern in more detail. In Table B.31, we classify a woman as a migrant if her place of residence is different from her birthplace, which could be too broad to capture migration flow in response to a contemporary local demand shock. Thus, we show alternative regression results when we code a person's migration status based on how many years she has lived in her location (Table B.32). We find no significant increase in migrants, regardless of when they arrived in the community. This result suggests that the composition of the communities is strongly stable in response to the shock. Given no major changes in the population composition, the first channel described above is less plausible. Therefore, the intimate partner violence could have been reduced via a "separation" effect due to less interaction between partners as they are not living together.

6.7 Alcohol Drinking

Studies on the link between alcohol consumption and domestic violence suggest that husband's or partner's alcohol drinking is a significant determinant of violent behavior and intimate partner violence in developed (Card and Dahl, 2011; Ivandić et al., 2024) and developing (Luca et al., 2015, 2019; Basu et al., 2025) countries. Since the employment and income from artisanal mining by husbands or male partners can increase their alcohol consumption through income effect[27] and high rates of binge drinking (Cawley and Ruhm, 2011) or reduce alcohol consumption via problem drinking effect (Mullahy and Sindelar, 1996). The DHS data collect information on whether a woman and her husband or partner drink alcohol. Leveraging this information, we first estimate the effect of artisanal gold mining on husband's alcohol drinking to examine if its impact on intimate partner violence is mediated through changes in men's alcohol consumption. Panel A of Table B.33 reports the results under four separate specifications wherein more controls are added successively. Our baseline and preferred specification in Column (4) suggests that the effect of artisanal gold mining on a husband or partner's alcohol consumption is essentially zero. This result provides evidence that ASgM does not reduce intimate partner violence via altering the husband's alcohol drinking and the associated behavioral changes.

Although the data on women's alcohol drinking status is available for limited countries and DHS rounds, we estimate the effect of artisanal gold mining on women's drinking behavior and find that ASgM reduces women's alcohol consumption. The impact is statistically significant at the 10% level (Panel B of Table B.33). This finding supports our results on the positive effect of ASgM on female employment, which is more significant than the male employment effect. Women reduce alcohol drinking as they supply their labor to broader industries in response to the positive economic shock. However, industrial gold mining strongly increases women's alcohol consumption, and the effect is larger both in magnitude and statistical significance. It potentially indicates the non-gender-inclusive feature of industrial mining.

6.8 Fertility Decision

So far, we have focused on explaining how ASgM might have reduced physical violence and thus overall domestic violence. However, the baseline effects on detailed forms of intimate partner violence suggest that ASgM increases a specific form of sexual violence, i.e., women being forced into unwanted sexual acts other than sex. Women's better labor market opportunities and earning

[27] See Fogarty (2010) for a comprehensive literature review on studies estimating the income elasticity of demand for alcoholic drinks.

potentials might make a wife postpone fertility (Baizán, 2007; Bloom et al., 2009; Aaronson et al., 2021; Doepke et al., 2023), while the husband wants a child. So we estimate the impact on fertility decisions proxied by the number of children, and results are shown in Table B.34. The effects of ASgM on different fertility measures are statistically insignificant; however, the signs of the coefficients are negative.

Although the impact is not statistically significant or imprecisely estimated, the result provides suggestive evidence on the negative effect of female labor force participation and fertility, which could explain an increase in the wife's experience of unwanted sexual acts by her husband or partner. The insignificant impact of ASgM on fertility, however, is consistent with the insignificant effect on overall female employment. It is also possible that the husband wanted sexual acts other than sex when the wife refused to have sex preventing them from having children.

7 Conclusion

This paper presents evidence that expansions of artisanal gold mining reduce intimate partner violence in Sub-Saharan Africa. Using nationally representative data from 23 countries, we provide population-based evidence on the relationship between mining activity and violence within the household. To identify the causal effects, we exploit spatial variation in gold suitability, the geospatial location of industrial gold mines, and exogenous temporal variation in global gold prices.

Our findings show that, in the short run, declines in less severe and less frequent physical violence drive the overall reduction in intimate partner violence. We find that women's intrahousehold bargaining power increases, suggesting that the bargaining power channel dominates potential male backlash effects in this context. Although overall employment does not change significantly, women's employment increases relative to men's in mining and retail industries, consistent with labor reallocation away from agriculture and local structural transformation. Together with evidence of rising household wealth, these patterns point to strengthened women's relative economic conditions. Overall, the findings are consistent with a framework in which improvements in women's economic opportunities and bargaining position reduce violence by intimate partners. However, the effects are heterogeneous across different forms of violence: while physical violence declines, some forms of sexual violence increase, and the long-run effects differ across violence categories. Importantly, we also show that artisanal and industrial mining have opposite effects on intimate partner violence, with industrial mining associated with increases in violence.

These findings have several implications. First, distinguishing between artisanal and industrial mining is important, as the two forms of mining operate through different labor market structures and generate significantly different social consequences. Second, improvements in women's economic opportunities can reduce domestic violence, with little evidence of dominant male backlash effects in this setting. Third, the heterogeneous effects across different forms of violence highlight the importance of nuanced policy approaches when evaluating the welfare consequences of resource booms.

Finally, we conclude with some caveats and directions for future research. While we document improvements in women's bargaining power and economic conditions, the available data capture labor market activity only on the extensive margin. Future work using richer labor market and earnings data could further examine intensive-margin adjustments and better isolate the mechanisms linking mining activity to household behavior. More broadly, combining granular spatial data on resource endowments with nationally representative micro data, as in this paper, offers a promising framework for studying the socio-economic consequences of extractive activities across a wide range of outcomes.

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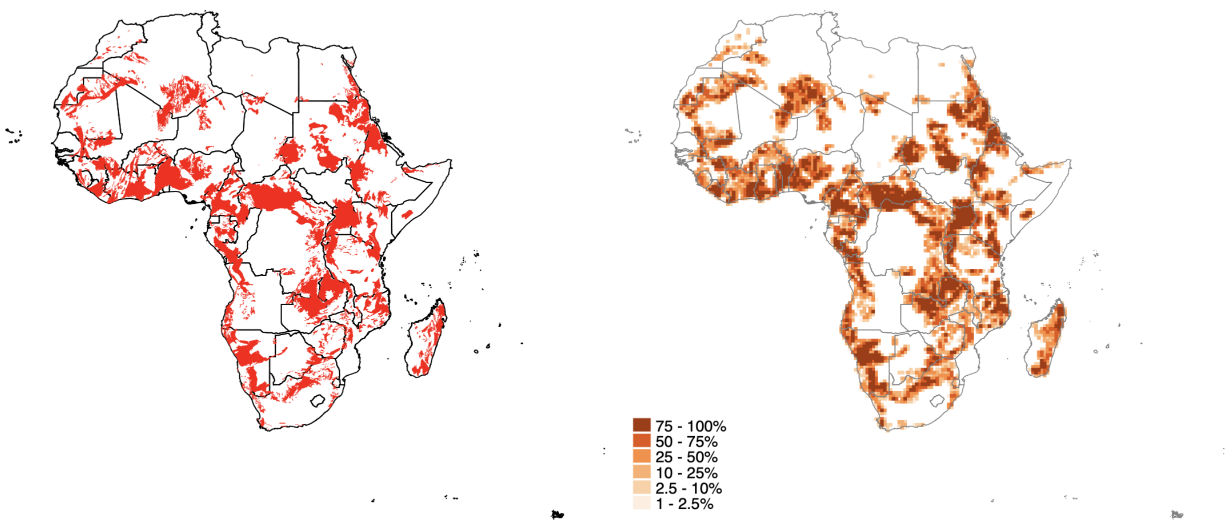
Figures

Figure 1: Artisanal Gold Mining in Africa

(a) Gold suitable bedrocks

(b) Share of the cell area suitable for gold

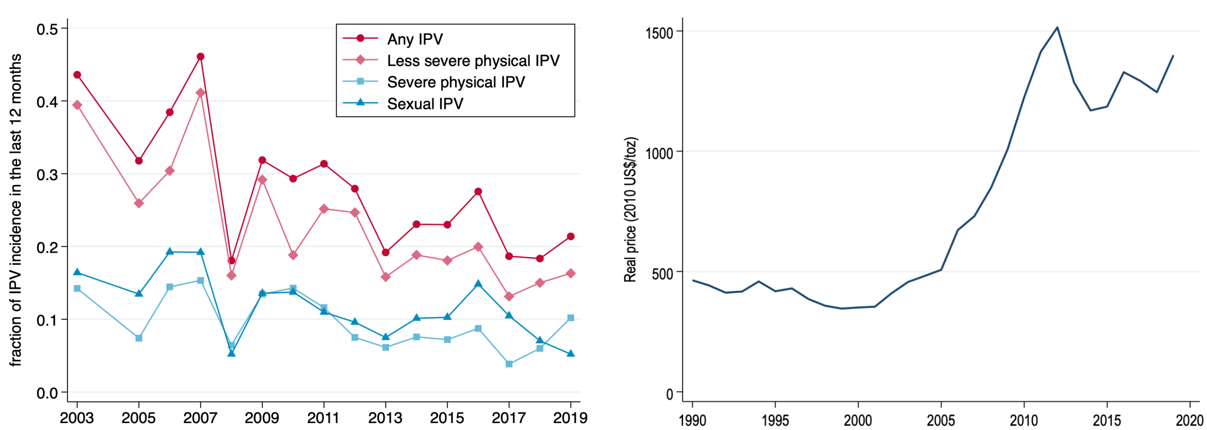


Notes: Panel (a) depicts the geological bedrocks suitable for artisanal gold mining. Panel (b) shows the percentage of areas suitable for artisanal gold mining in the PRIO-GRID cells of 0.5×0.5 degrees or 55×55 kilometers at the Equator.

Figure 2: Trends of Domestic Violence and Global Gold Price

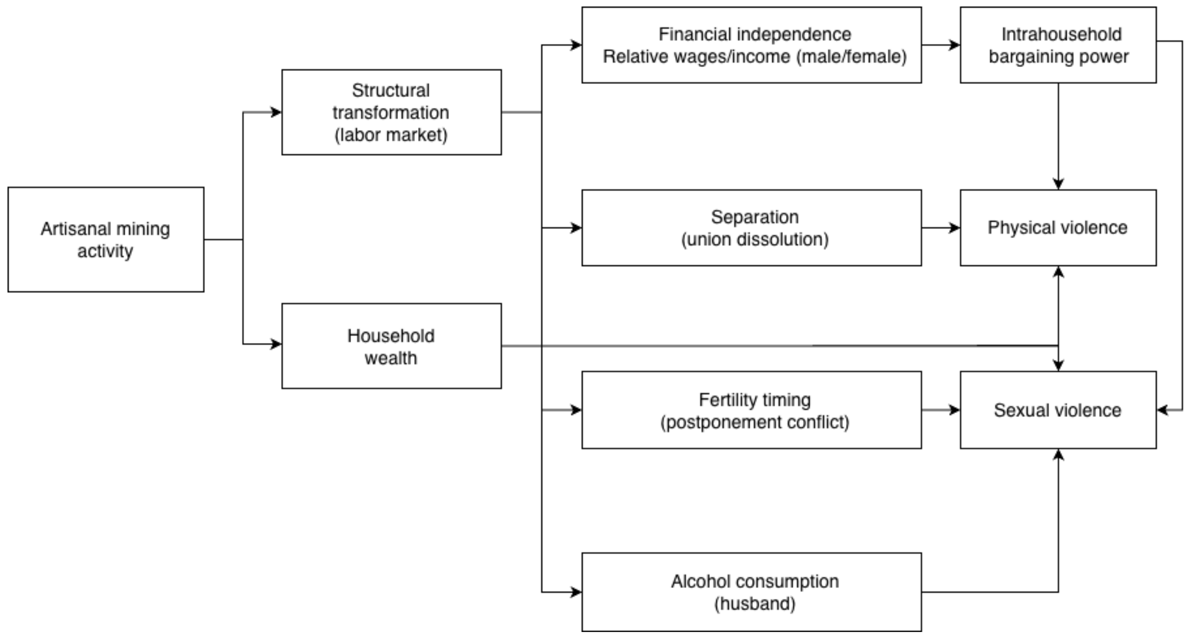
(a) Domestic Violence, 2003-2019

(b) Global gold price, 1990-2019



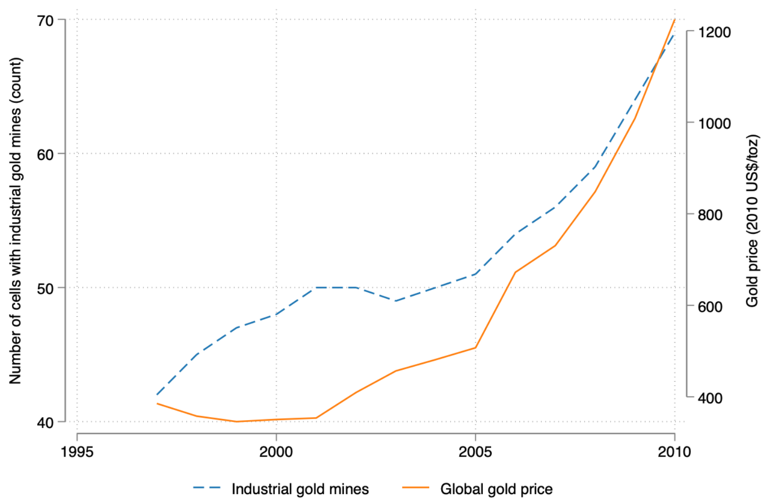
Notes: Panel (a) plots the trend of intimate partner violence in Sub-Saharan Africa using the DHS data from 2003-2019. The domestic violence data was unavailable in 2004 for all countries in our sample. The sample weight is applied to calculate the average incidence of intimate partner violence. Panel (b) shows the annual average price of gold (in 2010 US$/troy ounce) in the international market between 1990 and 2019 using data from the World Bank's Pink Sheet.

Figure 3: Summary of Conceptual Framework



Notes: The figure presents the possible mechanisms through which artisanal mining can affect intimate partner violence.

Figure 4: Relationship between Global Gold Price and Industrial Gold Mines



Notes: The figure presents the trends of industrial gold mines in Africa and global gold price between 1997 and 2010.

Tables

Table 1: Effects of Artisanal and Industrial Gold Mining on Broad Forms of Domestic Violence

	Dependent variable: A dummy variable for domestic violence			
	(1) Overall violence	(2) Less severe physical violence	(3) Severe physical violence	(4) Sexual violence
	Panel A. Ever experienced			
Gold suitable × Gold price	-0.109***	-0.095**	-0.023	-0.045
	(0.039)	(0.039)	(0.034)	(0.045)
Industrial gold mines × Gold price	0.131***	0.173***	0.018	0.050
	(0.049)	(0.059)	(0.051)	(0.062)
N	69990	69987	69979	69972
R^2	0.18	0.17	0.12	0.13
	Panel B. Sometimes experienced			
Gold suitable × Gold price	-0.120***	-0.100**	-0.051	-0.029
	(0.043)	(0.043)	(0.031)	(0.039)
Industrial gold mines × Gold price	0.159***	0.174***	0.061	0.054
	(0.058)	(0.064)	(0.059)	(0.050)
N	69813	68275	69478	68665
R^2	0.21	0.17	0.14	0.12
	Panel C. Often experienced			
Gold suitable × Gold price	-0.009	-0.011	0.010	-0.022
	(0.029)	(0.025)	(0.018)	(0.028)
Industrial gold mines × Gold price	-0.089*	-0.076	-0.064*	-0.006
	(0.052)	(0.053)	(0.036)	(0.042)
N	69840	67703	69250	67516
R^2	0.20	0.21	0.09	0.11

Notes: The table presents the OLS estimates on the effects of artisanal gold mining on overall and different forms of intimate partner violence. The dependent variables are dummy variables if a woman ever (Panel A), sometimes (Panel B), and often (Panel C) experienced broad forms of domestic violence in the last twelve months. Experiences of overall, less severe physical, severe physical, and sexual violence are considered in Columns (1)-(4), respectively. The key explanatory variable is our baseline measure of artisanal gold mining (an interaction of the proportion of the cell's surface suitable for gold with the lagged value of the international gold price). The effects of industrial gold mining are also presented, and the measure of industrial gold mining is our baseline measure (an interaction of a dummy indicating whether a cell has industrial gold mines with the lagged value of the international gold price). All regressions include individual and household characteristics, other cell-level covariates, cell fixed effects (FEs), and country-by-year FEs. The individual or woman characteristics include age, education level, current marital status, and religion. The household characteristics include urban/rural status and household size. The cell-level covariates include agricultural potential and weather conditions. The unit of observation is the woman. Standard errors clustered by cells are in parentheses. Significance: $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table 2: Heterogeneous Effects on Broad Forms of Domestic Violence by 3G Coverage

	Dependent variable: A dummy variable for domestic violence			
	(1) Overall violence	(2) Less severe physical violence	(3) Severe physical violence	(4) Sexual violence
	Panel A. Ever experienced			
Gold suitable × Gold price × 3G	-0.744**	-0.771**	-0.208	-0.373*
	(0.297)	(0.307)	(0.238)	(0.216)
Gold suitable × Gold price	-0.077*	-0.050	-0.053	-0.047
	(0.040)	(0.040)	(0.037)	(0.055)
N	69990	69987	69979	69972
R^2	0.18	0.17	0.12	0.13
	Panel B. Sometimes experienced			
Gold suitable × Gold price × 3G	-0.517	-0.550*	0.064	-0.342*
	(0.324)	(0.327)	(0.181)	(0.200)
Gold suitable × Gold price	-0.074*	-0.039	-0.072**	-0.014
	(0.044)	(0.045)	(0.034)	(0.047)
N	69813	68275	69478	68665
R^2	0.21	0.17	0.14	0.12
	Panel C. Often experienced			
Gold suitable × Gold price × 3G	-0.407**	-0.269*	-0.198	-0.220*
	(0.181)	(0.158)	(0.131)	(0.112)
Gold suitable × Gold price	-0.016	-0.013	0.010	-0.028
	(0.034)	(0.030)	(0.021)	(0.035)
N	69840	67703	69250	67516
R^2	0.20	0.21	0.09	0.11

Notes: The table presents the OLS estimates on the effects of artisanal gold mining on overall and different forms of intimate partner violence heterogeneous by 3G coverage. The dependent variables are dummy variables if a woman ever (Panel A), sometimes (Panel B), and often (Panel C) experienced broad forms of domestic violence in the last twelve months. Experiences of overall, less severe physical, severe physical, and sexual violence are considered in Columns (1)-(4), respectively. The key explanatory variable is a triple interaction between the proportion of the cell's surface suitable for gold, the lagged value of the international gold price, and the lagged dummy for a cell with 3G coverage. All regressions include the double interaction terms, baseline individual and household characteristics, cell-level covariates, and fixed effects. The unit of observation is the woman. Standard errors clustered by cells are in parentheses. Significance: *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 3: Effects on Women's Intrahousehold Bargaining Power Outcomes

	(1) Aggregate index 1	(2) Aggregate index 2	(3) Own money	(4) Own health	(5) Large purchase	(6) Daily purchase	(7) Visit	(8) Cook	(9) Husband money
Gold suitable	0.049***	0.063***	0.015	0.035	0.056**	0.094***	0.098***	0.036	-0.009
$\times$ Gold price	(0.016)	(0.020)	(0.018)	(0.024)	(0.023)	(0.034)	(0.027)	(0.040)	(0.038)
N	553906	546237	263824	541890	546110	211684	546097	129996	421505
R^2	0.35	0.34	0.13	0.27	0.26	0.28	0.24	0.20	0.25

Notes: The table presents the effects of artisanal gold mining on women's intrahousehold bargaining power using OLS regressions. The outcomes are dummy variables, indicating if the woman has the final say, individually or jointly with her husband/partner/someone else, on (i) spending her own money, (ii) her health care, (iii) making large purchases, (iv) making daily purchases, (v) making family visits, (vi) deciding what to cook daily, and (vii) spending her husband's money. The aggregate index 1 is calculated by taking a simple average of all the dummies, and aggregate index 2 is a simple average of the three most commonly asked questions in the DHS: (ii), (iii), and (v). The key explanatory variable is the interaction between gold suitability share and one-year lagged gold price index. All regressions include the baseline demographic and environmental controls and fixed effects. The unit of observation is the individual. Standard errors clustered by cells are in parentheses. Significance: *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$.

Table 4: Employment Effects of Artisanal Gold Mining

	Dependent variable: A dummy variable for employment				
	(1) All	(2) Mining	(3) Agriculture	(4) Services	(5) Sales
	Panel A. Overall				
Gold suitable × Gold price	-0.002	0.009*	-0.022	-0.028**	0.027
	(0.006)	(0.005)	(0.017)	(0.012)	(0.020)
Industrial gold mines × Gold price	0.007	0.001	-0.043	-0.012	0.055**
	(0.012)	(0.010)	(0.031)	(0.016)	(0.025)
N	634408	486671	615923	609080	609080
R^2	0.09	0.08	0.42	0.11	0.18
	Panel B. Male				
Gold suitable × Gold price	-0.002	0.005	-0.025	-0.024**	0.004
	(0.005)	(0.004)	(0.015)	(0.009)	(0.010)
Industrial gold mines × Gold price	0.016	-0.001	-0.039	-0.001	0.023
	(0.013)	(0.009)	(0.031)	(0.014)	(0.016)
N	606214	464848	588112	572860	572860
R^2	0.08	0.08	0.35	0.08	0.09
	Panel C. Female				
Gold suitable × Gold price	-0.021	0.008***	-0.033	-0.014	0.036
	(0.021)	(0.002)	(0.021)	(0.011)	(0.023)
Industrial gold mines × Gold price	0.018	0.003	-0.075*	-0.016	0.047
	(0.028)	(0.004)	(0.040)	(0.020)	(0.039)
N	631259	352085	436722	437355	437355
R^2	0.20	0.06	0.44	0.13	0.27

Notes: The table presents the effects of artisanal gold mining on employment or labor supply using OLS regressions. The dependent variables are dummy variables indicating whether an individual is employed in different industries. The aggregate, male, and female employment are considered in Panels A-C, respectively. Employment outcomes in Columns (1)-(5) are activities in all industries, mining, agriculture, services, and sales, respectively, in the last twelve months. The key explanatory variable is our baseline measure of artisanal gold mining (an interaction of the proportion of the cell's surface suitable for gold with the lagged value of the international gold price). All regressions include the baseline demographic and environmental controls and fixed effects. The unit of observation is the individual. Standard errors clustered by cells are in parentheses. Significance: $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.

Table 5: Effects of Artisanal Gold Mining on Household Wealth

	Dependent variable: Household wealth index, standardized				
	(1)	(2)	(3)	(4)	(5)
Gold suitable × Gold price (lag 1)	0.091*				0.088*
	(0.047)				(0.047)
Gold suitable × Log price		0.070**			
		(0.033)			
Gold suitable × Current price			0.099*		
			(0.053)		
Gold suitable dummy × Gold price				0.073**	
				(0.034)	
Industrial gold mines × Gold price	0.123*	0.120*	0.125*	0.118*	
	(0.068)	(0.068)	(0.068)	(0.070)	
Industrial mining × Mineral price					0.139**
					(0.058)
Crop suitable × Crop price	-0.002	-0.001	-0.000	-0.005	-0.002
	(0.043)	(0.043)	(0.043)	(0.042)	(0.043)
Temperature	-0.188***	-0.189***	-0.189***	-0.187***	-0.187***
	(0.038)	(0.038)	(0.038)	(0.038)	(0.038)
N	495843	495843	495843	495843	497914
R^2	0.63	0.63	0.63	0.63	0.63

Notes: The table presents the effects of artisanal gold mining on household wealth using OLS regressions. The dependent variable is the standardized household wealth index. The key explanatory variable is an alternative specification of an interaction of a cell's gold suitability with the international gold price. Column (1) is a baseline specification, an interaction of gold suitability share with a one-year lagged gold price index. Column (2) uses log gold price. Column (3) uses contemporaneous price. Column (4) uses a gold suitability dummy. Column (5) controls industrial mineral mining. All regressions include the baseline demographic and environmental controls and fixed effects. The unit of observation is the household. Standard errors clustered by cells are in parentheses. Significance: $*p < 0.10$, $**p < 0.05$, and $***p < 0.01$.